

Analyse I/ Fiche de TD2
Nombres complexes

Exercice 1 : Forme algébrique et calculs directs

1. Donner la forme algébrique de :

$$u_1 = (3 + 2i)(5 - i), \quad u_2 = (1 - i)^3, \quad u_3 = \frac{2 + 3i}{1 - i}$$

2. Simplifier :

$$u_4 = i^{2025} + i^{2026}$$

Exercice 2 : Module, argument et formes trigonométriques

1. Calculer le module et un argument de :

$$z_1 = -2 + 2i, \quad z_2 = \sqrt{3} - i, \quad z_3 = \frac{1 + i\sqrt{3}}{1 - i}$$

2. Soit

$$z = \frac{1 + i \tan \theta}{1 - i \tan \theta} \text{ avec } \theta \in \left] -\frac{\pi}{2}, \frac{\pi}{2} \right[.$$

Montrer que $|z| = 1$ et déterminer un argument de z en fonction de θ .

Exercice 3 : On considère les nombres complexes suivants :

$$z_1 = 1 + i\sqrt{3}, \quad z_2 = 1 + i, \quad z_3 = \frac{z_1}{z_2}$$

1. Écrire z_3 sous forme algébrique.
2. Écrire z_3 sous forme trigonométrique.
3. En déduire les valeurs exactes de $\cos \frac{\pi}{12}$ et $\sin \frac{\pi}{12}$.

Exercice 4 : Forme exponentielle et puissances

1. Mettre sous forme exponentielle :

$$w_1 = 4i, \quad w_2 = -3\sqrt{2} - 3i\sqrt{2}, \quad w_3 = \frac{1}{2} - i\frac{\sqrt{3}}{2}$$

2. Soit $j = e^{i2\pi/3}$. Calculer $(1 + j)^{2024}$ sous forme algébrique.

Exercice 5 : Racines et équations du second degré

1. Déterminer les racines carrées de $-2i$, $5 - 12i$.
2. Résoudre dans $\mathbb{C}$:

$$z^2 - (3 - i)z + 4 - 3i = 0 ; \quad z^2 - 2mz + m^2 + 4 = 0, \quad m \in \mathbb{R}.$$

3. Trouver les racines cubiques de $8i$.

Exo1 :

Couragé de la fiche TD2 (nombres complexes)

$$1) U_1 = (3+2i)(5-i) = 15+7i+2 = 17+7i$$

$$U_2 = (1-i)^3 = (1-i)(-2i) = -2i-2$$

$$U_3 = \frac{2+3i}{1-i} = \frac{(2+3i)(1+i)}{2} = \frac{-1+5i}{2}$$

$$2) U_4 = i^{2025} + i^{2026}$$

$$\text{Orma: } i^4 = 1 \text{ et } 2025 = 4 \times 506 + 1$$

$$2026 = 4 \times 506 + 2$$

$$\text{donc } U_4 = i^{4 \times 506 + 1} + i^{4 \times 506 + 2}$$

$$= 1^{506} \cdot i + 1^{506} \cdot i^2 = i - 1 = -1 + i$$

$$U_5 = \frac{(1+i)^{10}}{(1-i)^8} =$$

Exo2 :

$$1) |z_1| = \sqrt{4+4} = 2\sqrt{2} \quad z_1 = -2+2i \quad \left. \begin{array}{l} \cos \alpha = \frac{-2}{2\sqrt{2}} = -\frac{\sqrt{2}}{2} \\ \sin \alpha = \frac{2}{2\sqrt{2}} = \frac{\sqrt{2}}{2} \end{array} \right\} \alpha = \frac{3\pi}{4} = \frac{\pi}{4} + \frac{\pi}{2} \quad \left. \begin{array}{l} \cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2} \\ \sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2} \end{array} \right\} z_1 = 2\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$$z_2 = \sqrt{3} - i \quad |z_2| = \sqrt{4} = 2 \quad \left. \begin{array}{l} \cos \alpha = \frac{\sqrt{3}}{2} \\ \sin \alpha = -\frac{1}{2} \end{array} \right\} \alpha = -\frac{\pi}{6} \quad \left. \begin{array}{l} \cos \left(-\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} \\ \sin \left(-\frac{\pi}{6}\right) = -\frac{1}{2} \end{array} \right\} z_2 = 2 \left(\cos \left(-\frac{\pi}{6}\right) + i \sin \left(-\frac{\pi}{6}\right) \right) = 2 \left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)$$

$$z_3 = \frac{1+i\sqrt{3}}{1-i} = \frac{z_4}{z_5} \quad / \quad z_4 = 1+i\sqrt{3} \quad , \quad z_5 = 1-i$$

$$|z_3| = \frac{|z_4|}{|z_5|} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$\arg z_3 = \arg z_4 - \arg z_5 = \frac{\pi}{3} - \left(-\frac{\pi}{4}\right) = \frac{7\pi}{12}$$

$$\left. \begin{array}{l} \cos \alpha = \frac{1}{2} \\ \sin \alpha = \frac{\sqrt{3}}{2} \end{array} \right\} \alpha = \frac{\pi}{3}$$

$$\left. \begin{array}{l} \cos \beta = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \\ \sin \beta = -\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2} \end{array} \right\} \beta = -\frac{\pi}{4}$$

$$z_3 = \sqrt{2} \left(\cos\left(\frac{7\pi}{12}\right) + i \sin\left(\frac{7\pi}{12}\right) \right)$$

$$2/ z = \frac{1+i \tan \theta}{1-i \tan \theta}$$

$$|z| = \frac{|1+i \tan \theta|}{|1-i \tan \theta|} = \frac{\sqrt{1+\tan^2 \theta}}{\sqrt{1+\tan^2 \theta}} = 1$$

$$\arg z = ?$$

1^{ère} Méthode: $z = \frac{1 + i \frac{\sin \theta}{\cos \theta}}{1 - i \frac{\sin \theta}{\cos \theta}} = \frac{\cos \theta + i \sin \theta}{\cos \theta - i \sin \theta}$

donc $z = \frac{(\cos \theta + i \sin \theta)(\cos \theta + i \sin \theta)}{\cos^2 \theta + \sin^2 \theta}$

$$z = \frac{(\cos^2 \theta - \sin^2 \theta) + i(2 \sin \theta \cos \theta)}{1}$$

$$z = \cos 2\theta + i \sin 2\theta \Rightarrow \arg z = 2\theta$$

2^{ème} Méthode:

$$z = \frac{\cos \theta + i \sin \theta}{\cos \theta - i \sin \theta} = \frac{\cos \theta + i \sin \theta}{\cos(-\theta) + i \sin(-\theta)}$$

$$\arg z = \arg(\cos \theta + i \sin \theta) - \arg(\cos(-\theta) + i \sin(-\theta))$$

$$= \theta - (-\theta) = 2\theta$$

Exo 11: 1) $w_1 = 4i$, $|w_1| = 4$

$$\left. \begin{array}{l} \cos \alpha = 0 \\ \sin \alpha = 1 \end{array} \right\} \alpha = \frac{\pi}{2} \Rightarrow w_1 = 4 e^{i\pi/2}$$

$$w_2 = -3\sqrt{2} + 3i\sqrt{2} \quad |w_2| = 3\sqrt{2+2} = 6$$

$$\left. \begin{array}{l} \cos \alpha = \frac{-3\sqrt{2}}{6} = -\frac{\sqrt{2}}{2} \\ \sin \alpha = \frac{3\sqrt{2}}{6} = \frac{\sqrt{2}}{2} \end{array} \right\} \alpha = \pi - \frac{\pi}{4} = \frac{3\pi}{4} \Rightarrow w_2 = 6 e^{i\frac{3\pi}{4}}$$

$$w_3 = \frac{1}{2} - i \frac{\sqrt{3}}{2} \quad |w_3| = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1$$

$$\left. \begin{aligned} \cos \alpha &= \frac{1}{2} \\ \sin \alpha &= -\frac{\sqrt{3}}{2} \end{aligned} \right\} \Rightarrow \alpha = -\frac{\pi}{3} \Rightarrow w_3 = e^{-i\frac{\pi}{3}}$$

$$2/j = e^{i2\pi/3} = \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) = -\frac{1}{2} + i \frac{\sqrt{3}}{2}$$

$$1+j = \frac{1}{2} + i \frac{\sqrt{3}}{2} = e^{i\frac{\pi}{3}}$$

$$\begin{aligned} (1+j)^{2024} &= e^{i \frac{2024}{3} \pi} = e^{i(674 \times 3 + 2) \frac{\pi}{3}} \\ &= e^{i(674\pi + \frac{2\pi}{3})} = e^{i \frac{2\pi}{3}} = j = -\frac{1}{2} + i \frac{\sqrt{3}}{2} \end{aligned}$$

Exo 5 :

1) Racine carrée: $z = -2i$

Cherchons $w = x + iy$ / $w^2 = z$.

$$\begin{aligned} w^2 = z \Leftrightarrow \begin{cases} x^2 - y^2 + 2ixy = -2i \\ |w|^2 = |z| \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 0 \dots (1) \\ xy = -1 \dots (2) \\ x^2 + y^2 = 2 \dots (3) \end{cases} \end{aligned}$$

$$(1) + (3) \Leftrightarrow x^2 = 1 \Rightarrow x = 1 \text{ ou } x = -1$$

$$\begin{aligned} \text{dans (2)} \quad x=1 &\Rightarrow y=-1 & \text{d'où } w_1 &= 1-i \\ x=-1 &\Rightarrow y=1 & w_2 &= -1+i \end{aligned}$$

ou bien on peut utiliser la forme exponentielle.

$$z = -2i = 2 e^{-i\frac{\pi}{2}} \quad \text{alors } w_1 = \sqrt{2} e^{-i\frac{\pi}{4}} \text{ ou } w_2 = -\sqrt{2} e^{-i\frac{\pi}{4}}$$

$$\star z = 5 - 12i$$

Cherchons $w = x + iy$ / $w^2 = z$.

$$\begin{aligned} w^2 = x^2 - y^2 + 2ixy = 5 - 12i \Leftrightarrow \begin{cases} x^2 - y^2 = 5 \dots (1) \\ 2xy = -12 \dots (2) \\ |w|^2 = |z| = \sqrt{25 + 144} = \sqrt{169} = 13 \dots (3) \end{cases} \end{aligned}$$

$$\textcircled{1} + \textcircled{3} \Rightarrow 2x^2 = 18 \Rightarrow x^2 = 9 \Rightarrow x = 3 \text{ ou } x = -3$$

dans $\textcircled{2}$: $xy = -6$; si $x = 3 \Rightarrow y = -2$

si $x = -3 \Rightarrow y = 2$.

d'où $w_1 = 3 - 2i$, $w_2 = -3 + 2i$ sont les 2 racines carrées de z .

$$2) z^2 - (3-i)z + 4-3i = 0.$$

$$\Delta = (3-i)^2 - 4(4-3i) = 9 - 1 - 6i - 16 + 12i = -8 + 6i$$

cherchons la racine carrée de Δ par la même méthode

Si dessus, on trouve $\sqrt{\Delta} = 1 + 3i$ ou $-1 - 3i$

$$\text{d'où } z_1 = \frac{3-i+1+3i}{2} = 2+i, z_2 = \frac{3-i-1-3i}{2} = 1-2i$$

$$3) \text{ Racine cubique de } 8i = z$$

cherchons $w = r e^{i\theta}$ / $w^3 = z = 8i = 8 e^{i\pi/2}$

$$w^3 = r^3 e^{i3\theta} = 8 e^{i\pi/2}$$

$$\text{d'où } \begin{cases} r^3 = 8 \\ 3\theta = \frac{\pi}{2} + 2k\pi, k=0,1,2. \end{cases}$$

$$\begin{cases} r = 2 \\ \theta = \frac{\pi}{6} + \frac{2k\pi}{3}, k=0,1,2. \end{cases}$$

Pour $k=0$; $\theta = \pi/6$ et $w_1 = 2 e^{i\pi/6}$.

pour $k=1$; $\theta = \frac{5\pi}{6}$ et $w_2 = 2 e^{i5\pi/6}$

pour $k=2$; $\theta = \frac{9\pi}{6} = \frac{3\pi}{2}$ et $w_3 = 2 e^{i3\pi/2}$.

(Faint handwritten notes and diagrams at the bottom of the page, including a small sketch of a unit circle with points marked.)

$$2) \quad z^2 - 2mz + m^2 + 4 = 0$$

$$\Delta = 4m^2 - 4m^2 - 16 = -16 = i^2 16$$

$$z_1 = \frac{2m + 4i}{2} = m + 2i$$

$$z_2 = \frac{2m - 4i}{2} = m - 2i$$

$$\text{Ex03: } z_1 = -1 + i\sqrt{3}, \quad z_2 = 1 + i, \quad z_3 = \frac{z_1}{z_2}$$

$$1) z_3 = \frac{1 + i\sqrt{3}}{1 + i} = \frac{(1 + i\sqrt{3})(1 - i)}{(1 + i)(1 - i)} = \frac{1 - i + i\sqrt{3} + \sqrt{3}}{2} = \frac{(1 + \sqrt{3}) + i(-1 + \sqrt{3})}{2}$$

$$= \left(\frac{1 + \sqrt{3}}{2}\right) + i\left(\frac{-1 + \sqrt{3}}{2}\right) \rightarrow \text{forme Algébrique.}$$

$$2) z_3 = |z_3| (\cos \arg z_3 + i \sin \arg z_3) \quad / \quad |z_3| = \frac{|z_1|}{|z_2|}, \quad \arg z_3 = \arg z_1 - \arg z_2$$

$$|z_3| = \frac{|1 + i\sqrt{3}|}{|1 + i|} = \frac{2}{\sqrt{2}} = \frac{\sqrt{2} \cdot \sqrt{2}}{\sqrt{2}} = \sqrt{2} \quad / \quad |z_1| = 2, \quad |z_2| = \sqrt{2}$$

$$\arg z_1 = ? \quad \begin{cases} \cos \theta = \frac{1}{2} \\ \sin \theta = \frac{\sqrt{3}}{2} \end{cases} \Rightarrow \theta = \frac{\pi}{3}$$

$$\arg z_2 = ? \quad \begin{cases} \cos \alpha = \frac{1}{\sqrt{2}} \\ \sin \alpha = \frac{1}{\sqrt{2}} \end{cases} \Rightarrow \alpha = \frac{\pi}{4}$$

$$\arg z_3 = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12} \quad \text{d'où } z_3 = \sqrt{2} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right)$$

3) En identifiant la forme Algébrique avec la forme trigonométrique, on obtient:

$$\begin{cases} \frac{1 + \sqrt{3}}{2} = \sqrt{2} \cos \frac{\pi}{12} \\ -\frac{1 + \sqrt{3}}{2} = \sqrt{2} \sin \frac{\pi}{12} \end{cases} \Rightarrow \begin{cases} \cos \frac{\pi}{12} = \frac{1 + \sqrt{3}}{2\sqrt{2}} \\ \sin \frac{\pi}{12} = -\frac{1 + \sqrt{3}}{2\sqrt{2}} \end{cases}$$